

FTP server configuration

Objective: Configure secure file transfer services using SFTP (port 22) and VSFTPD (port 21) with restricted user access and verification using FTP clients like FileZilla or Cyberduck.

Description

GOAL: Enable file transaction to servers using file transfer protocol[FTP]

1. configure sftp in an instance such that, it can receive files via port number 22 to the specific directory as the specific user.ie, the sftp users have restricted privileges.
2. verify the working with any FTP client like filezilla, cyberduck etc.
3. install and configure vsftp in an instance such that, it can receive files very securely via port 21 to the configured directory as configures user.for Vsftp also privileges are restricted.
4. verify the working with any FTP client.

Part 1 — SFTP Configuration

1. Install and Enable OpenSSH Server

```
dnf install -y openssh-serversystemctl enable sshdsystemctl start sshd
```

2. Create Dedicated SFTP Group

```
groupadd sftpusers
```

3. Create Restrictive SFTP User Account

```
useradd -m -G sftpusers sftpuserpasswd sftpuser
```

4. Establish Isolated Chroot Directory Structure

```
mkdir -p /sftp/sftpuser/uploads
```

5. Enforce Strict Ownership and Access Permissions

```
chown root:root /sftpchmod 755 /sftpchown sftpuser:sftpuser /sftp/sftpuser/uploads
```

6. Restrict User Environment in SSH Configuration

```
Match Group sftpusers
    ChrootDirectory /sftp
    ForceCommand internal-sftp
    X11Forwarding no
    AllowTcpForwarding no
```

7. Apply Configuration Changes and Update Firewall

```
systemctl restart sshd firewall-cmd --permanent --add-service=sshd --reload
```

SFTP Verification (FileZilla / Cyberduck)

- **Protocol:** SFTP
- **Port:** 22
- **Credentials:** Enter server IP, username and password.

Part 2 — VSFTPD Configuration

1. Install and Bootstrap VSFTPD Daemon

```
dnf install -y vsftpd systemctl enable vsftpd systemctl start vsftpd
```

2. Create Dedicated FTP User Account

```
useradd ftpuserpasswd ftpuser
```

3. Provisions Custom Storage Hierarchy

```
mkdir -p /ftpdata/ftpuser/files
```

4. Assign Directory Permissions

```
chown -R ftpuser:ftpuser /ftpdata/ftpuserchmod -R 755 /ftpdata
```

5. Backup and Modify Core Configuration File

```
cp /etc/vsftpd/vsftpd.conf /etc/vsftpd/vsftpd.conf.bak
```

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
```

```
chroot_local_user=YES
pasv_enable=YES
```

6. Enforce Networking and Daemon Runtime Changes

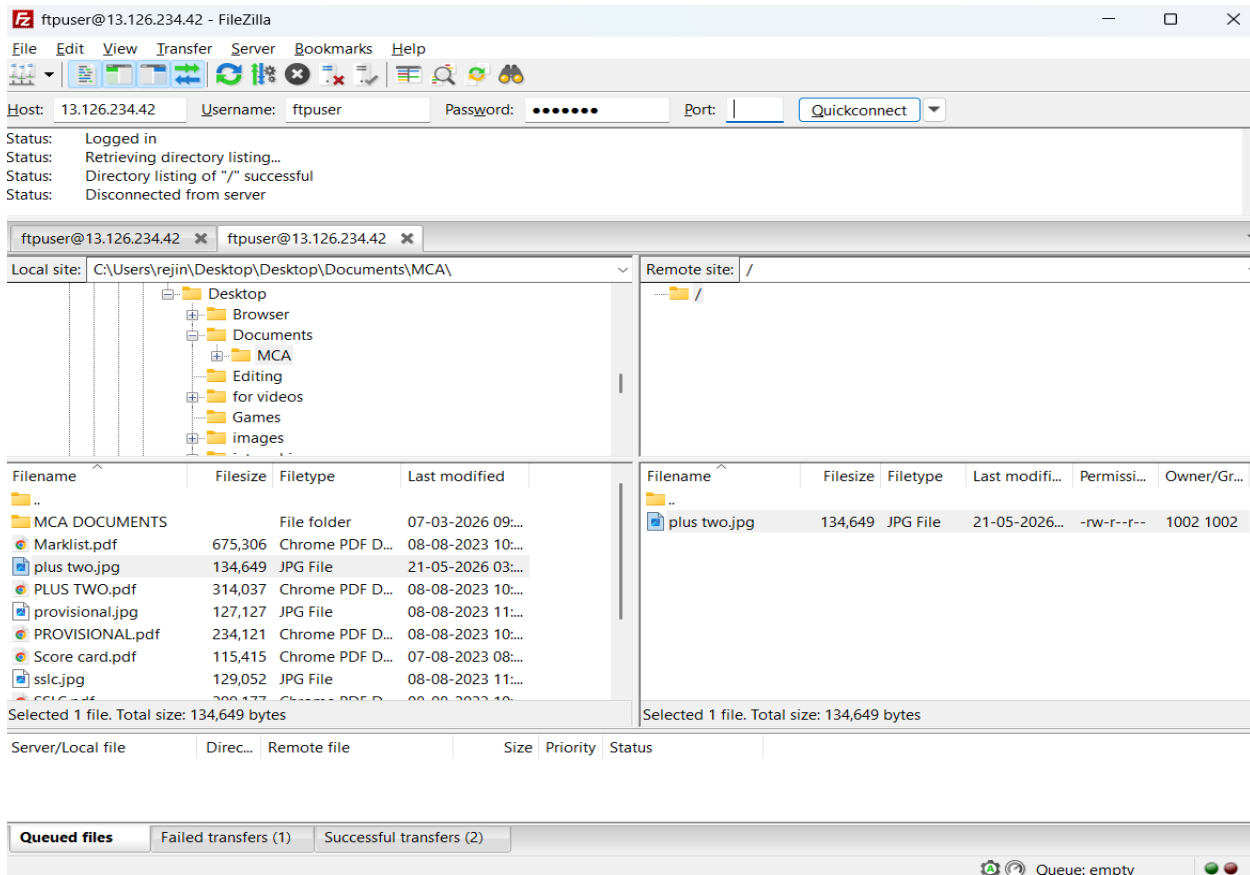
```
systemctl restart vsftpd
firewall-cmd --permanent --add-service=ftp
firewall-cmd --permanent --add-port=30000-31000/tcp
firewall-cmd --reload
```

FTP Verification (FileZilla / Cyberduck)

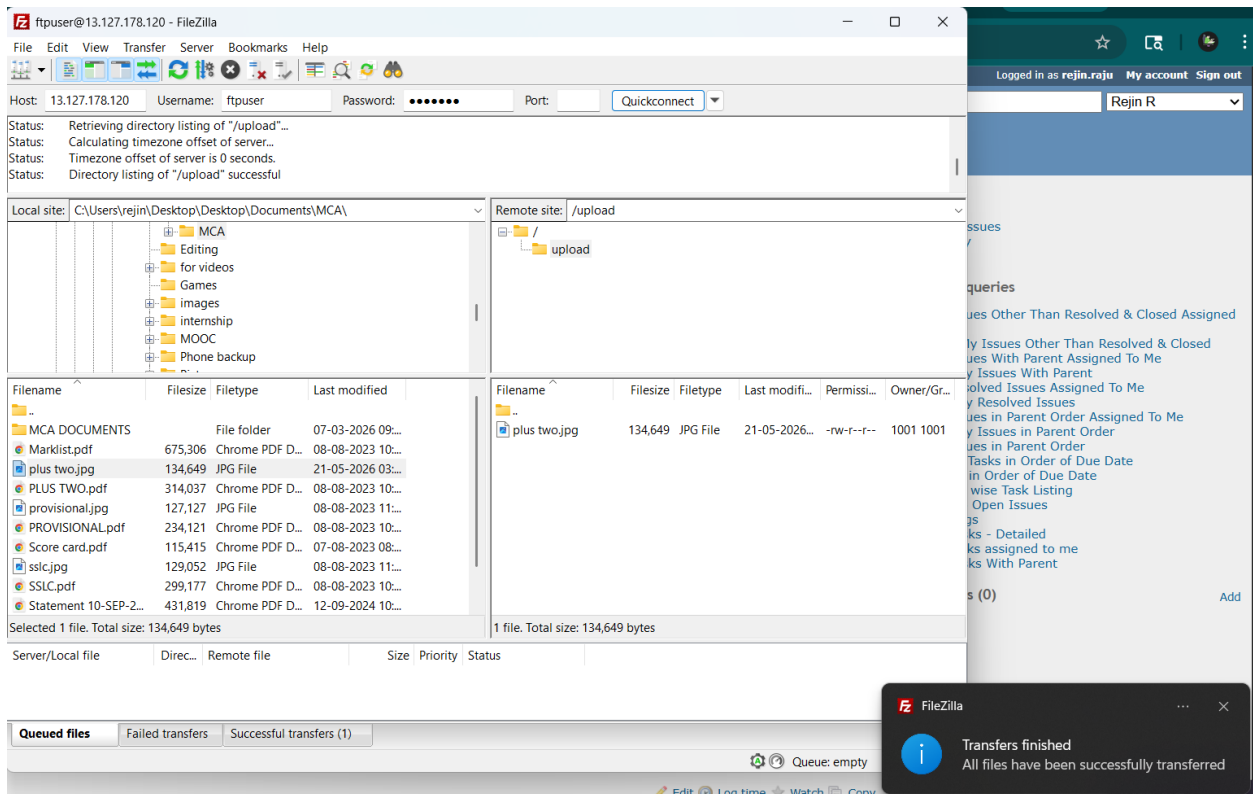
- **Protocol:** FTP
- **Port:** 21
- **Encryption:** Only use plain FTP (insecure).

Output Pictures

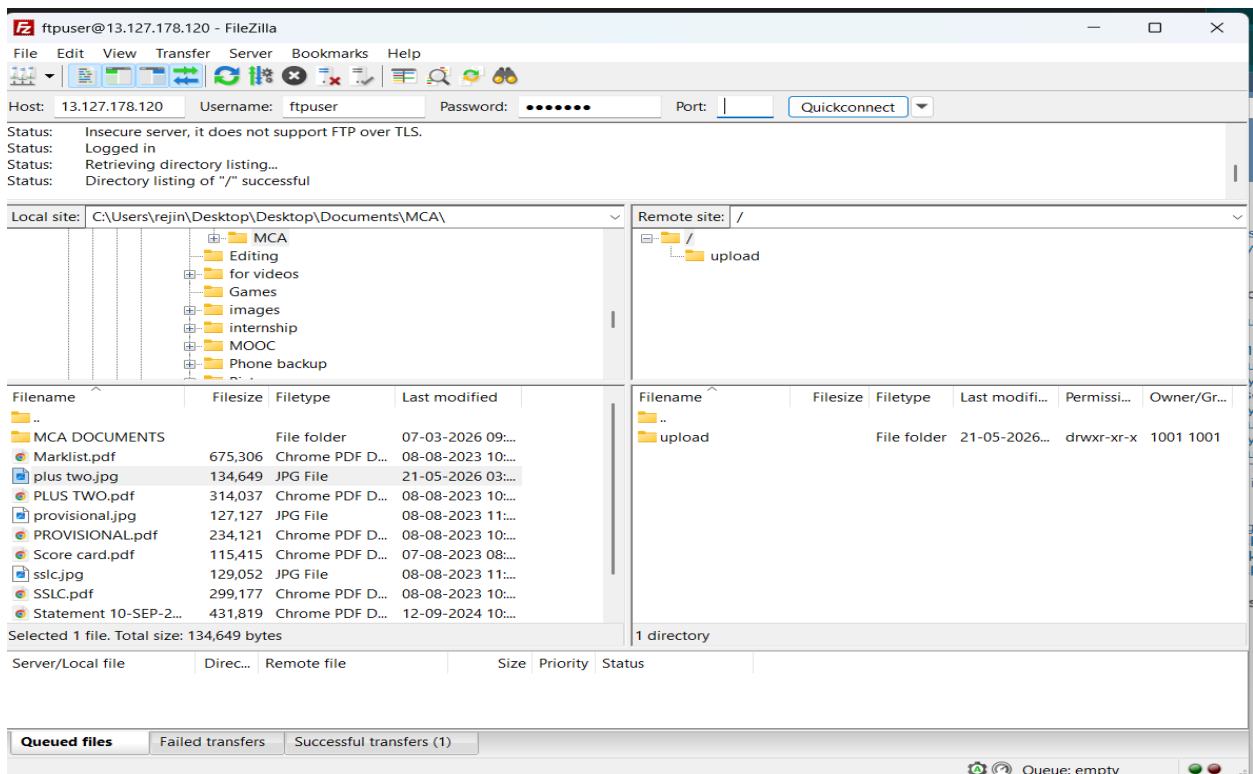
SFTP login result



Transfer successful picture



VSFTP login result



Transfer successful picture

sftp://rejin@13.126.234.42 - FileZilla

File Edit View Transfer Server Bookmarks Help

Host: sftp://13.126.234.42 Username: rejin Password: Port: Quickconnect

Status: Starting upload of C:\Users\rejin\Desktop\Desktop\Documents\MCA\provisional.jpg
Status: File transfer successful, transferred 127,127 bytes in 1 second
Status: Retrieving directory listing of "/upload"..
Status: Directory listing of "/upload" successful

Local site: C:\Users\rejin\Desktop\Desktop\Documents\MCA\ Remote site: /upload

MCA

Editing

for videos

Games

images

internship

MOOC

Phone backup

Filename	Filesize	Filetype	Last modified
MCA DOCUMENTS		File folder	07-03-2026 09:...
Marklist.pdf	675,306	Chrome PDF D...	08-08-2023 10:...
plus two.jpg	134,649	JPG File	21-05-2026 03:...
PLUS TWO.pdf	314,037	Chrome PDF D...	08-08-2023 10:...
provisional.jpg	127,127	JPG File	08-08-2023 11:...
PROVISIONAL.pdf	234,121	Chrome PDF D...	08-08-2023 10:...
Score card.pdf	115,415	Chrome PDF D...	07-08-2023 08:...
ssl.jpg	129,052	JPG File	08-08-2023 11:...
SSL.pdf	299,177	Chrome PDF D...	08-08-2023 10:...
Statement 10-SEP-2...	431,819	Chrome PDF D...	12-09-2024 10:...

/

upload

Filename	Filesize	Filetype	Last modified	Permissi...	Owner/Gr...
plus two.jpg	134,649	JPG File	21-05-2026...	-rw-r--r--	1001 1001
provisional.jpg	127,127	JPG File	21-05-2026...	-rw-r--r--	1001 1001

Selected 1 file. Total size: 127,127 bytes

2 files. Total size: 261,776 bytes

Server/Local file	Dir...	Remote file	Size	Priority	Status
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Queued files

Failed transfers

Successful transfers (2)

Queue: empty

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Rejin_r (668848431270)

Rejin_r

Launch instances

Public IPv4 DNS

ec2-13-126-234-42.ap-... 1

ec2-13-127-178-120.ap-... 1

Transfers finished

All files have been successfully transferred